

Cybersecurity Threats and Trends

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1)

Introduction

1.1 Types of Cybersecurity [1,2] (1/2)

- Endpoint security
 - Protects **end users and endpoint devices** (desktops, laptops, mobile devices, smartphones, servers) against cyberattacks. Organizations are also adopting unified endpoint management (UEM) solutions that allow them to protect, configure and manage all endpoint devices from a single console.
- Operational Technology-Critical Infrastructure security
 - Protects the computer systems, applications, networks, data and digital assets that a society depends on for national security, economic health and public safety.
- Identity security
 - Protects **digital identities** and the systems that manage them. It includes practices such as identity verification, access control enforcement and unauthorized access prevention.
- Information security-Data security
 - InfoSec protects an organization's important information (digital files and data, paper documents, physical media) against unauthorized access, use or alteration. Data security, the protection of digital information, is a subset of information security and the focus of most cybersecurity-related InfoSec measures.
- Network security
 - Prevents unauthorized access to computer networks and systems, detects and stops in progress cyberattacks and security breaches, and ensures that authorized users have secure access to their network resources.

1.1 Types of Cybersecurity [1,2] (2/2)

► Cloud security

- Secures an organization's **cloud-based infrastructure**, including applications, data and virtual servers. Generally, cloud security operates on the shared responsibility model. The cloud provider is responsible for securing their delivered services and the infrastructure that delivers them. The customer is responsible for protecting customer data, code and other assets they store or run in the cloud.

► Application security

- AppSec works to identify and repair vulnerabilities in application software to prevent unauthorized access, modification or misuse.

► AI security

- Protects AI applications and systems from cyberthreats, cyberattacks and malicious use.

► Zero-trust security

- A modern cybersecurity model that assumes no user or system, whether inside or outside the network, is automatically trustworthy by default. Instead, organizations continuously verify access to data and resources through **strict authentication protocols**.

1.2 Trends driving cyber threats [1,3]

- Cloud computing
 - On-demand access to computing resources can increase network management complexity and raise the risk of cloud misconfigurations, improperly secured application programming interfaces (APIs) and other avenues hackers can exploit.
- Multicloud environments
 - A multicloud approach introduces risks such as increased attack surface, identity access management gaps, cloud sprawl, fragmented security capabilities and heightened risk of human error or misconfiguration.
- Distributed-Remote work
 - Remote work, hybrid work and **bring-your-own-device** (BYOD) policies mean more connections, devices, applications and data, for security teams to protect and for threat actors to exploit.
- The Internet of Things (IoT)
 - Many connected devices (vehicles, appliances and other physical objects) within IoT networks are unsecured or improperly secured by default and can be easily hijacked by bad actors.
- Artificial Intelligence
 - **Generative AI** in particular is a new threat landscape already exploited through methods such as prompt injection. Research from the IBM Institute for Business Value says only 24% of generative AI initiatives are secured.



2)

Cyber Threats

2.1 Types of Cyber Threats

- Social Engineering--Phishing attacks
- Malware--Ransomware
- Network Attacks--Distributed Denial-of-Service (DDoS)
- Web and Application Attacks
- Cloud and Infrastructure Attacks
- Insider and Identity-based Attacks
- Advanced Attacks

2.1.1 Social Engineering [4] (1/2)

Social engineering attacks exploit human trust rather than technical flaws, **persuading people to reveal information or even install malware**. Phishing attacks are digital or voice messages that try to manipulate recipients into sharing sensitive information, downloading malicious software, transferring money or assets to the wrong people or taking some other damaging actions. Scammers craft phishing messages to look or sound like they come from a trusted or credible organization or individual, sometimes, even an individual the recipient knows personally.

- Bulk phishing emails
 - Emails, texts or social media messages mimic legitimate requests and lure victims into **clicking malicious links or opening infected attachments**.
- Spear phishing
 - Tailors the attack to a specific individual using details from public social profiles.
- Whale phishing
 - Spear phishing version aimed at senior executives.
- Business email compromise (BEC)
 - Scams **impersonate trusted individuals** like a CEO, tricking employees into wiring funds or sharing confidential data (type of spear phishing).
- Voice phishing (Vishing)
 - Phishing that is conducted through phone calls. Individuals typically experience vishing in the form of threatening recorded calls claiming to be from the FBI.

2.1.1 Social Engineering [4] (2/2)

- Angler phishing
 - Phishing using **fake social media accounts** that masquerade as the official accounts of trusted companies' customer service or customer support teams.
- Baiting
 - Lures victims into knowingly or unwittingly giving up sensitive information or downloading malicious code by **tempting them with a valuable offer** or even a valuable object. The Nigerian Prince scam is probably the best-known example of this social engineering technique. More current examples include free but malware-infected games, music or software downloads. Some threat actors leave malware-infected USB drives where people will find them, grab them and use them because they are free.
- Tailgating-piggybacking
 - An unauthorized person closely follows an authorized person into an area containing sensitive information or valuable assets. Tailgating can be conducted in person, for example, a threat actor can follow an employee through an unlocked door. But tailgating can also be a digital tactic, such as **when a person leaves a computer unattended** while still logged in to a private account or network.
- Pretexting
 - The threat actor **creates a fake situation for the victim** and poses as the right person to resolve it. Very often (and most ironically) the scammer claims that the victim has been impacted by a security breach and then offers to fix things if the victim will provide important account information or control over the victim's computer or device. Technically speaking, almost every social engineering attack involves some degree of pretexting.
- Quid pro quo
 - Hackers dangle a desirable good or service **in exchange for the victim's sensitive information**. Fake contest winnings or seemingly innocent loyalty rewards are examples of quid pro quo plays.

2.1.2 Malware [4] (1/2)

Malware is **malicious software** that can render infected systems inoperable. It can destroy data, steal information or wipe files critical to the operating system's ability to run. Common types of malware include:

- Trojan horses
 - Attacks **masquerade as legitimate programs** to trick users into installation. A remote access Trojan (RAT) opens a secret back door on the victim's device, while a dropper Trojan installs additional malware after gaining a foothold.
- Ransomware
 - Uses strong **encryption** to hold data or systems hostage until a ransom is paid.
- Scareware
 - Bombards victims with **fake warnings** to induce downloads or the surrender of sensitive information.
- Spyware
 - **Secretly collects** usernames, passwords and credit card numbers, sending them back to the attacker.
- Rootkits
 - **Grant administrator-level control** of an operating system while remaining hidden.
- Self-replicating worms
 - Spread automatically between applications and devices.

2.1.2 Malware (2/2)

- Backdoors
- Rogue software
- Keylogging
- Fileless attacks
 - Use vulnerabilities in legitimate software programs to inject malicious code directly into a computer's memory. Because they operate only in memory and leave few artifacts on disk, they can evade many antivirus software solutions, even some next-generation antivirus (NGAV) tools. Attackers often leverage scripting environments like PowerShell to change configurations or steal passwords. [4]
- Malvertising

2.1.3 Network Attacks [4] (1/2)

- Denial-of-service (DoS)
 - Attacks **flood a system's resources** with fraudulent traffic until it can't respond to legitimate requests. A DoS attack originates from a single source.
- Distributed denial-of-service (DDoS)
 - Uses multiple sources, often a botnet of malware-infected laptops, smartphones and Internet-of-Things (IoT) devices.
- Man-in-the-middle attacks (MITM)--eavesdropping
 - The hacker secretly **intercepts communications** between two parties, often over unsecured public Wi-Fi. Attackers can read or modify messages before they reach the recipient. For instance, in a session-hijacking variant, the intruder swaps their IP address with the victim's, fooling the server into granting full access to protected resources.
- Domain name system (DNS) tunneling
 - **Hides malicious traffic inside DNS** packets, allowing it to bypass traditional security measures such as firewalls and intrusion detection systems (IDS). Threat actors use this technique to create covert communication channels that can silently extract data or connect malware to a remote command and control (C2) server.
- DNS spoofing-poisoning
 - Covertly alters DNS records to **replace a website's real IP address** with a fraudulent one. When victims attempt to visit the legitimate site, they are unknowingly redirected to a malicious copy that can steal data or distribute malware.

2.1.3 Network Attacks (2/2)

- URL interpretation
- Session Hijacking
- Drive-by attack
- Eavesdropping
- Network sniffing

2.1.4 Web and Application Attacks [4] (1/2)

- Cross-site scripting (XSS)
 - Inserts malicious code into a legitimate web page or web application. When a user visits the site or app, the code automatically runs in the user's browser, stealing sensitive information or redirecting the visitor to a malicious website. Attackers frequently use JavaScript to launch these exploits.
- SQL injection
- Zero-day exploits
 - Take advantage of previously unknown or unpatched **software flaws** known as zero-day vulnerabilities before developers can release a fix. These attacks can remain effective for days, months or even years, making them a favorite of advanced threat groups.

2.1.4 Web and Application Attacks (2/2)

- Click jacking
- Web attack
- Birthday
- Form jacking
- Code injection

2.1.5 Cloud and Infrastructure Attacks

- Supply chain attacks
 - Attackers breach a company by targeting its software vendors, material suppliers or other service providers. Because **vendors** are frequently connected to their customers' networks, a single compromise can provide an attacker with an indirect path into many organizations. [4]
- IoT based
 - An IoT-based attack is a cyberattack that exploits **vulnerabilities in internet-connected devices** to gain unauthorized access, steal data, or cause damage. Common attacks include DDoS, malware, credential theft and man-in-the-middle attacks, often enabled by weak security on devices like default passwords, outdated firmware, and insecure interfaces.
- Cloud and container exploits
 - Enterprises continue to shift workloads to public and hybrid clouds, expanding the potential **attack surface**. Misconfigured storage buckets, exposed application programming interfaces (APIs) and vulnerable container-orchestration platforms like Kubernetes give attackers opportunities to gain access to entire environments in near real time. Targeting a single cloud misconfiguration can let a threat actor **move laterally across multiple workloads** and exfiltrate customer data without triggering traditional perimeter defenses. [4]

2.1.6 Insider and Identity-based Attacks

[1] (1/2)

- Insider threats
 - Originate with authorized users (employees, contractors or business partners) who intentionally or accidentally **misuse their legitimate access** or have their accounts hijacked by cybercriminals. These threats can be difficult to detect because they have the earmarks of authorized activity and are invisible to antivirus software, firewalls and other security solutions that block external attacks.
- Credential theft and account abuse
 - Hackers have many techniques for stealing credentials and taking over accounts. For example, Kerberoasting attacks manipulate the Kerberos authentication protocol (commonly used in Microsoft Active Directory) to seize privileged service accounts. In 2025, the IBM X-Force team identified a surge in phishing emails distributing infostealer malware and credential phishing.
- Account compromise
 - Criminals **hijack a legitimate user's credentials** to conduct malicious activity. They may phish for passwords, purchase stolen databases on the dark web or launch automated brute force attacks to repetitively guess passwords until one works. [4]

2.1.6 Insider and Identity-based Attacks (2/2)

- Identity-based
- Password attacks
- Brute force attacks
- Dictionary attacks

2.1.7 Advanced Attacks [4]

- AI driven attacks
 - Cybercriminals are using AI to conduct advanced attacks. Some use open source generative AI to produce fake emails, applications and other business documents in minutes. Hackers are also using organizations' AI tools as attack vectors. For example, in prompt injection attacks, threat actors use malicious inputs to manipulate generative AI systems into leaking sensitive data, spreading misinformation or worse.
- Data tampering
 - Aims to corrupt or subtly **alter datasets**, whether in transit, in storage or during processing, so that downstream systems make flawed decisions. This can include manipulating real-time data streams or quietly editing financial or healthcare records. One particularly serious tactic is data poisoning, in which attackers modify machine learning training sets with malicious records, causing models to develop hidden backdoors or biased outputs.
- Quantum-era risks
 - Advances in quantum computing **threaten today's public-key cryptography**. Attackers are already pursuing "harvest now, decrypt later" strategies, stealing encrypted data today with the expectation that future quantum capabilities will allow them to break current encryption algorithms and unlock sensitive information. Preparing for this shift requires organizations to track developments in post-quantum cryptography (PQC) and begin planning migration paths for critical systems.
- Cryptojacking
 - Cryptojacking occurs when hackers gain access to a device and use its computing resources to **mine cryptocurrencies** such as Bitcoin, Ethereum and Monero. Security analysts identified cryptojacking as a cyberthreat around 2011, shortly after the introduction of cryptocurrency.
- Watering hole

2.2 OWASP Top 10 for 2021 [5] (1/3)

The OWASP Top 10 provides a concise overview of the most prevalent security risks facing web applications. By highlighting these risks, developers can proactively incorporate appropriate security measures into their applications, reducing the likelihood of security breaches and unauthorized access.

➤ Broken Access Control

- Access control enforces policy such that users cannot act outside of their intended permissions. Failures typically lead to unauthorized information disclosure, modification and destruction, or performing a business function outside the user's limits.

➤ Cryptographic Failure

- What to do: determine the protection needs of data in transit and at rest. For example, passwords, credit card numbers, health records, personal information and business secrets require extra protection, mainly if that data falls under privacy laws. For all such data ensure that **strong cryptographic algorithms** are used.

➤ Injection

- Some of the more common injections are SQL, NoSQL, OS command, Object Relational Mapping (ORM), LDAP, Expression Language (EL) and Object Graph Navigation Library (OGNL) injection. **Source code review** is the best method of detecting if applications are vulnerable to injections. Automated testing of all parameters, headers, URL, cookies, JSON, SOAP, and XML data inputs is strongly encouraged.

➤ Insecure Design

- An insecure design cannot be fixed by a perfect implementation, as the security controls were never created to defend against specific attacks. One of the factors that contribute to insecure design is the lack of business risk profiling inherent in the software or system being developed, and thus the failure to determine what level of security design is required.

2.2 OWASP Top 10 for 2021 [5] (2/3)

➤ Security Misconfiguration

- The application might be vulnerable if: the application is missing appropriate **security hardening**, unnecessary features are enabled or installed, default accounts and their passwords are still enabled and unchanged, error handling reveals stack traces or other overly informative error messages to users, for upgraded systems the latest security features are disabled or not configured securely, the **security settings** are not set to secure values, the server does not send security headers or directives, the software is **out of date** or vulnerable.

➤ Vulnerable and Outdated Components

- You are likely vulnerable: if you do not know the **versions** of all components you use, if the software is vulnerable, unsupported, or out of date, if you do not **scan for vulnerabilities** regularly, if you do not fix or upgrade the underlying platform, frameworks, and dependencies in a risk-based, timely fashion, if software developers do not test the compatibility of updated, upgraded, or patched libraries, if you do not secure the components' configurations.

➤ Identification and Authentication Failures

- There may be authentication weaknesses if the application: permits **automated attacks**, permits brute force attacks, permits default, weak, or well-known **passwords**, uses weak or ineffective **credential recovery** and forgot-password processes, uses plain text or weakly hashed passwords data stores, has missing or ineffective multi-factor authentication, exposes session identifier in the URL, reuses **session identifier** after successful login, does not correctly invalidate Session IDs, user sessions or authentication tokens aren't properly invalidated during logout or a period of inactivity.

2.2 OWASP Top 10 for 2021 [5] (3/3)

■ Software and Data Integrity Failures

- Relate to code and infrastructure that does not protect against integrity violations. An example of this is where an application relies upon plugins, libraries, or modules from **untrusted sources**, repositories, and content delivery networks (CDNs). An insecure continuous integration/continuous development (CI/CD) pipeline can introduce the potential for unauthorized access, malicious code, or system compromise. Lastly, many applications now include **auto-update** functionality, where updates are downloaded without sufficient integrity verification and applied to the previously trusted application. Attackers could potentially upload their own updates to be distributed and run on all installations. Another example is where objects or data are encoded or serialized into a structure that an attacker can see and modify.

■ Security Logging and Monitoring Failures

- Without logging and monitoring, **breaches cannot be detected**. Insufficient logging, detection, monitoring, and active response occurs any time: auditable events are not logged, warnings and errors generate no, inadequate, or unclear **log messages**, logs of applications and APIs are not monitored for suspicious activity, logs are only stored locally, appropriate alerting thresholds and response escalation processes are not in place or effective, penetration testing and scans by dynamic application security testing tools do not **trigger alerts**, the application cannot detect, escalate, or alert for active attacks in **real-time** or near real-time. Vulnerability to information leakage occurs by making logging and alerting events **visible** to a user or an attacker. Vulnerability to injections or attacks on the logging or monitoring systems occurs if log data is not correctly **encoded**.

■ Server-Side Request Forgery

- Occurs whenever a web application is fetching a remote resource without **validating the user-supplied URL**. It allows an attacker to coerce the application to send a crafted request to an unexpected destination, even when protected by a firewall, VPN, or another type of network access control list (ACL). As modern web applications provide end-users with convenient features, fetching a URL becomes a common scenario. As a result, the incidence of SSRF is increasing. Also, the severity of SSRF is becoming higher due to cloud services and the complexity of architectures.



3)

Cyber Threats 2025

3.1 ENISA Threat Landscape

- ENISA Threat Landscape: overview of the most prominent cybersecurity threats and trends the EU faces in the current cyber threat ecosystem.
- Threat-centric approach
- Threat-informed defence involves people, tools and technologies, processes, decision-making.

3.2 Most identified initial infection vector [6]

- Phishing (60%)
- Vulnerabilities (21.3%)
- Botnet (9.9%)
- Malicious applications (8%)
- Insider threat (0.8%)

3.3 Incident types [6]

- The distribution of incident types is dominated by DDoS attacks, which make up about 76.7% of recorded cases. This category is overwhelmingly driven by hacktivist groups, which account for most collected DDoS incidents, with cybercrime groups contributing a marginal fraction. Intrusions follow with 17.8%, dominated by cybercriminal activities, followed by state aligned intrusion sets, which typically pursue persistence. Hacktivists appear only marginally in intrusion cases. Defacements were almost exclusively associated with hacktivists, underlining their role as a symbolic tactic for visibility and protest rather than a sustained intrusion method.
- Distributed Denial of Service (76.7%)
- Intrusion (17.8%)
- Unknown (5%)
- Unauthorized access (0.4%)
- Defacement (0.1%)

3.4 Distribution of identified malicious codes [6]

- The prevalence of cybercriminal-led intrusions is illustrated through the type of malicious code deployed following intrusions, as well as the outcome of recorded intrusions.
- Ransomware (83.5%)
- Backdoor (5%)
- Spyware (4%)
- Stealer (2.7%)
- RAT (2.7%)
- Banking trojan (1.2%)
- Webshell (0.7%)
- Keylogger (0.3%)

3.5 Distribution of threat categories [6]

- Mobile threats (42.4%)
- Web threats (27.3%)
- Operational technology (18.2%)
- Supply chain (10.6%)
- Exposure (1.5%)

3.6 Targeted sectors in the EU [6]

- Over the reporting period, ENISA collected and curated 4875 events. 28.5% of the total number of incidents were not associated to a specific sector, either because the sector was not properly documented (i.e., private sector, private companies) or not mentioned at all.
- Public administration (38.2%)
- Transport (7.5%)
- Digital infrastructure and services (4.8%)
- Finance (4.5%)
- Manufacturing (2.9%)
- Media and entertainment (2.3%)
- Business services (2.2%)
- Energy (1.7%)
- Education (1.7%)
- Health (1.2%)



4)

TTPs and Vulnerabilities

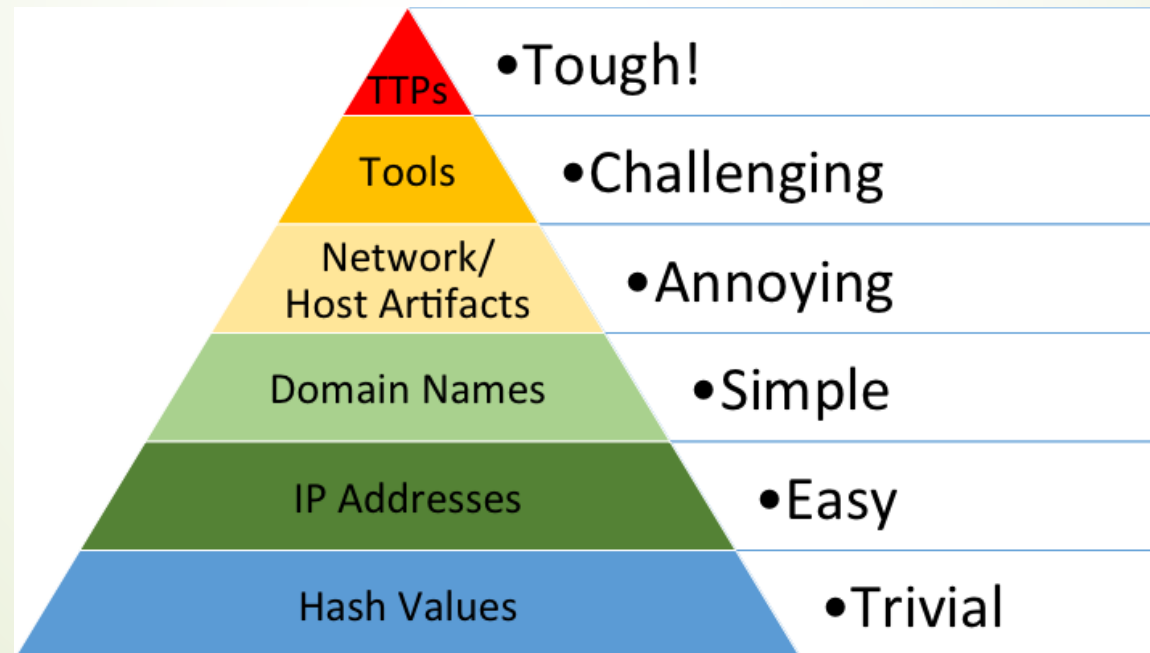
4.1 Tactics, Techniques and Procedures (TTPs)

- The behavior of an actor. A tactic is the highest-level description of this behavior, while techniques give a more detailed description of behavior in the context of a tactic, and procedures an even lower-level, highly detailed description in the context of a technique. [7]
- ATT&CK is knowledge base of adversarial techniques based on real-world observations. ATT&CK focuses on how adversaries interact with systems during an operation, reflecting the various phases of an adversary's attack lifecycle and the platforms they are known to target. The main components of the model are:
 - Tactics that represent “why” or the reason an adversary is performing an action.
 - Techniques that represent “how” adversaries achieve tactical goals by performing an action.
 - Sub-techniques which is a more specific or lower-level description of adversarial behavior.
 - Procedures that give specific implementation of techniques or sub-techniques.[8]
- ATT&CK is organized in three technology domains: enterprise, mobile and industrial control systems (ICS).
- Presentation on how to use ATT&CK and cooperate with other employees effectively:

<https://i.blackhat.com/USA-19/Wednesday/us-19-Nickels-MITRE-ATTACK-The-Play-At-Home-Edition.pdf> [9]

4.2 Pyramid of Pain [10]

- This diagram shows the relationship between the types of indicators one might use to detect an adversary's activities and how much pain it will cause them when they are able to deny those indicators to them.

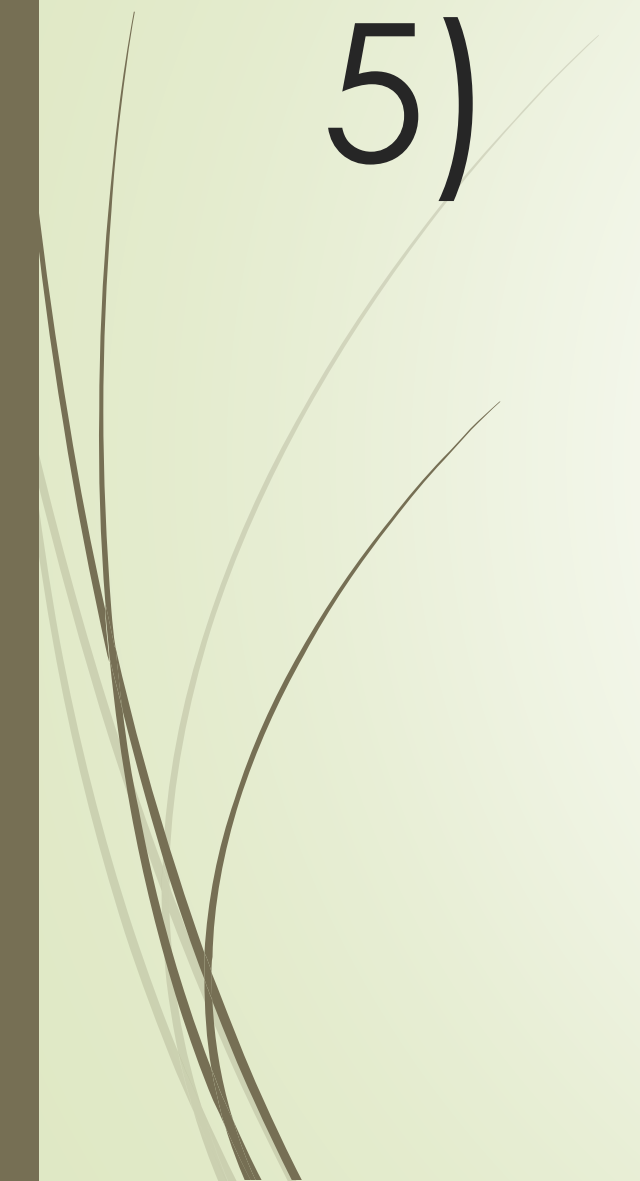


4.2 Vulnerabilities

- An instance of one or more weaknesses in a product that can be exploited, causing a negative impact to confidentiality, integrity, or availability. A set of conditions or behaviors that allows the violation of an explicit or implicit security policy. [11]
- European Union Vulnerability Database provides reliable and timely information related to vulnerabilities. [12]



5)



Cybersecurity Organizations

5. Cybersecurity related Organizations

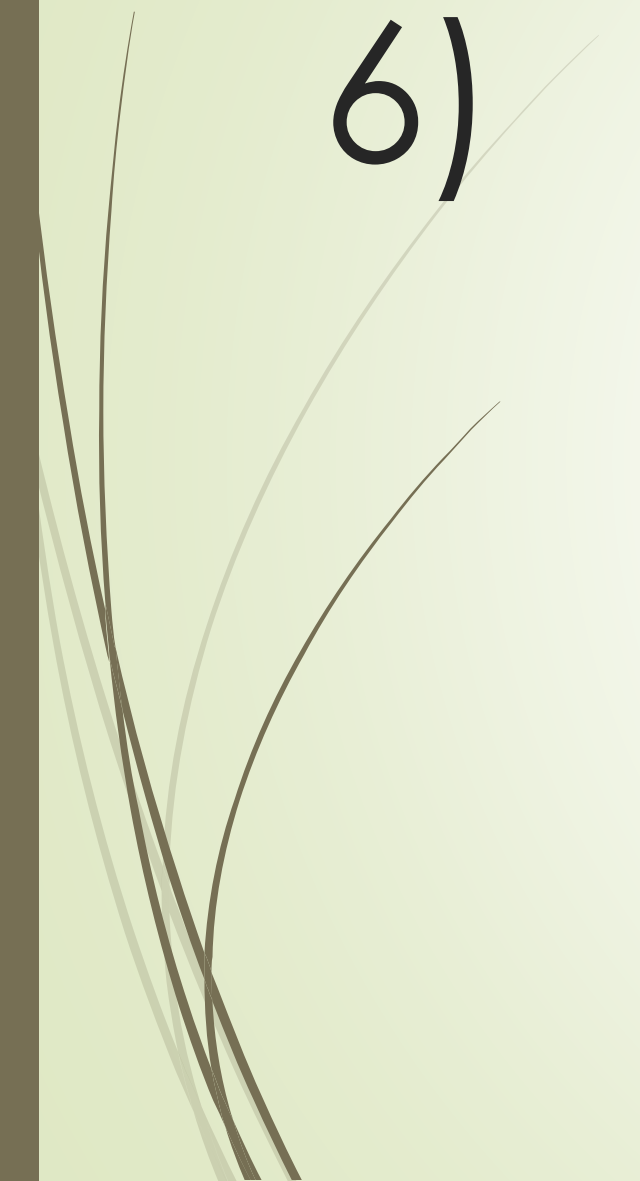
- NIST = National Institute of Standards and Technology
- ENISA = European Union Agency for Cybersecurity
- CSA = Cloud Security Alliance
- SAFECode = Software Assurance Forum for Excellence in Code
- MITRE ATT&CK
- OWASP = Open Web Application Security Project
- CIS = Center for Internet Security
- CISA = Cybersecurity and Infrastructure Security Agency

5. NIST/ENISA

- NIST=National Institute of Standards and Technology
 - NIST Cybersecurity Framework (CSF) is a voluntary set of guidelines that helps organizations manage and reduce cybersecurity risks by providing a structured approach to best practices in five core functions: Identify, Protect, Detect, Respond, and Recover. NIST also provides resources on various cybersecurity topics, including passwords, incident response, and cryptography. [13]
- ENISA = European Union Agency for Cybersecurity
 - ENISA serves as the European Union's center of expertise in cybersecurity. Its primary goal is to enhance cybersecurity resilience across Europe by developing policy guidance, supporting the protection of critical infrastructure, and strengthening cooperation among EU Member States, businesses, and citizens. The agency promotes trust in digital systems through initiatives such as awareness programs, knowledge sharing, and the development of EU-wide cybersecurity certification schemes. [14]



6)



Acronyms and Terminology

6.1 Acronyms (1/2)

- ▶ UEM = Unified End-point Management
- ▶ AI = Artificial Intelligence
- ▶ API = Application Programming Interface
- ▶ BYOD = bring-your-own-device
- ▶ IoT = Internet of Things
- ▶ DDoS = Distributed Denial-of-Service
- ▶ BEC = Business Email Compromise
- ▶ RAT = Remote Access Trojan
- ▶ NGAV = Next-Generation Antivirus
- ▶ MITM = man-in-the-middle
- ▶ DNS = Domain Name System
- ▶ IDS = Intrusion Detection System
- ▶ URL = Uniform Resource Locator

6.1 Acronyms (2/2)

- XSS = Cross-site Scripting
- SQL = Structured Query Language
- PQC = Post-Quantum Cryptography
- LDAP = Lightweight Directory Access Protocol
- JSON = JavaScript Object Notation
- JWT = JSON Web Token
- CDN = Content Delivery Network
- CI/CD = Continuous Integration/Continuous Development
- IP = Internet Protocol
- TTPs = Tactics, Techniques and Procedures
- ICS = Industrial Control Systems
- CSF = Cybersecurity Framework

6.2 Terminology (1/2)

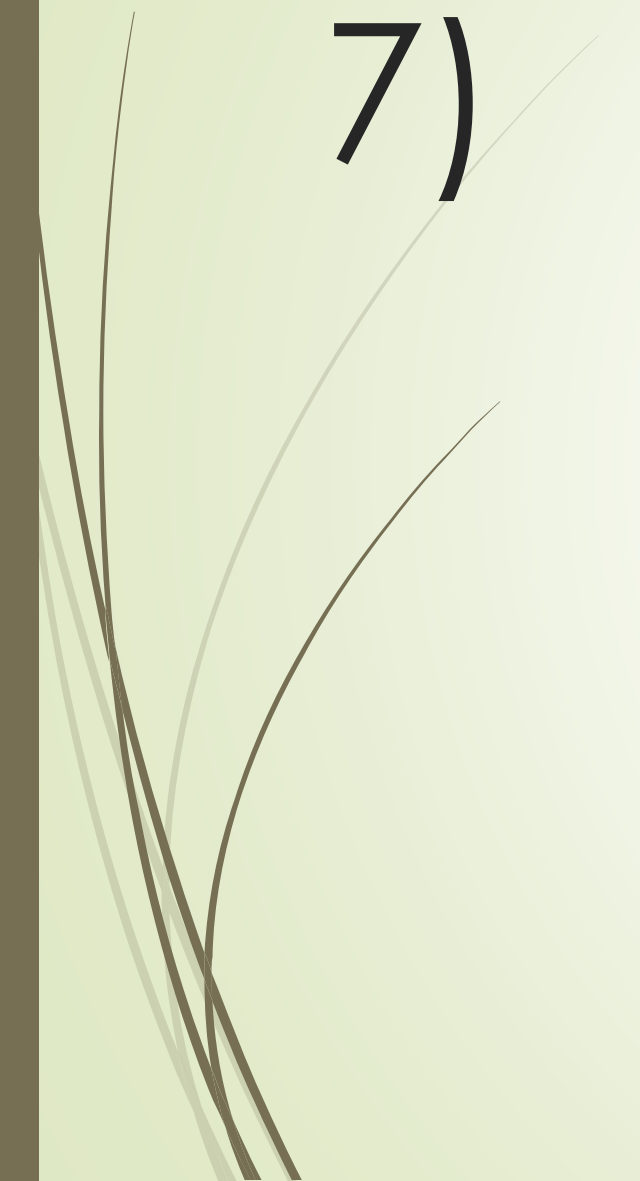
- Hacktivism
- Lateral movement
- Threat intelligence
- Threat emulation

6.2 Terminology (2/2)

- **Hacktivism** is the use of computer-based techniques such as hacking as a form of civil disobedience to promote a political agenda or social change. A form of internet activism with roots in hacker culture and hacker ethics, its ends are often related to free speech, human rights, or freedom of information movements.
- **Lateral movement** refers to the techniques that a cyberattacker uses, after gaining initial access, to move deeper into a network in search of sensitive data and other high-value assets. After entering the network, the attacker maintains ongoing access by moving through the compromised environment and obtaining increased privileges using various tools. [15]
- **Threat intelligence** refers to the collection, processing, and analysis of data to understand a threat actor's motives, targets, and attack methods. It transforms raw data into actionable insights, enabling security teams to make informed, data-driven decisions. This shifts organizations from a reactive to a proactive stance in defending against cyber threats. [16]
- **Threat emulation** is an intelligence-driven impersonation of real-world attack scenarios and TTPs in a controlled environment to test, assess and improve an organisation's security defences and response capabilities. This means that one seeks to behave as the adversary would. Threat emulation aims to identify and mitigate security gaps before attackers exploit them. [17]



7)



References and Licenses

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Take-home message

No matter how informative or well-structured a presentation may be, it can't replace personal initiative and independent research. Take charge of your learning: look up a new word you heard today, ask someone about it, or follow a trusted source for insightful content. Small, consistent steps lead to deeper learning and lasting self-growth.